

Feature Neutralization

Explained + Residual

Model predictions with one feature can be written as a straight line - fitted part. The left-over is the residual.

$$\text{prediction} = \text{fitted} + \text{residual}$$

Feature exposure is what measures the fitted part.
Feature neutralization is what throws away some of it.

Linear regression

We can find best line to explain predictions with one feature with linear regression: $\text{fitted} = \beta \times \text{feature}$

With one feature the best slope β is $\beta = \text{cov}(X, Y) / \text{var}(X)$
With many features we need to use matrix pseudo-inverse.

Proportion knob

We can decide how much of fitted part we want to remove with a multiplier, ending with this formula:

$$\text{neutralized} = \text{prediction} - \text{proportion} \times \text{fitted}$$